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What's in a villain's name?

Sound symbolic values of voiced obstruents and bilabial consonants

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This paper reports two case studies of sound symbolism using the naturalistic name corpora of characters from Disney and Pokémon. Building upon previous studies of sound symbolism, we tested two hypotheses: (1) voiced obstruents, which are generally associated with negative images, are favored in villainous characters' names, while (2) bilabial consonants, which are symbolically associated with cuteness, are disfavored in such names. The results show that these tendencies hold in our corpora, suggesting that a concept that is as complex as "villain" can be signaled via sound symbolism. Theoretical implications for cognitive linguistics based on our results are discussed.

Keywords: sound symbolism, evil/good, naming, obstruents, bilabials

1. Introduction

1.1 Theoretical background

This paper reports two case studies of sound symbolism, addressing the general question of whether sounds and meanings have systematic associations. Whether sounds have direct relationships with meanings or not has been a topic of intensive debate among philosophers, psychologists and linguists since antiquity, at least going back to Plato's *Cratylus* written during the classical period of Ancient Greece. However, since Saussure (1916/1972), it has been generally taken for granted in modern linguistic theories that relationships between sounds and meanings are fundamentally arbitrary. This is to say that, for example, there is

no intrinsic relationship between the sound sequence [k^hæt] and the animal that this sound sequence refers to in English. Indeed, the same animal can be called by different names in different languages; e.g., [neko] in Japanese, [katz] in German and [gato] in Spanish. As Shakespeare puts it, “a rose by any other name would smell as sweet.” In the same vein, there is no fundamental meaning associated with the sound [k] or [æ], for instance; generally, in this view, sounds in and of themselves are void of any meanings, and only when they are combined to be a word/morpheme do they bear a meaning.

While the arbitrariness between sounds and meanings is indeed one distinguishing property of human languages (Hockett, 1959), it has been recurrently observed that there are cases in which sounds and meanings have systematic relationships, patterns sometimes referred to “sound symbolism” (for recent reviews of sound symbolism research, see e.g., Dingemanse et al., 2015; Hinton et al., 2006; Lockwood & Dingemanse, 2015; Perniss et al., 2010; Sidhu & Pexman, 2017). Hinton et al. (2006, p.1), one of the landmark studies of sound symbolism in the modern era, define sound symbolism as “the direct linkage between sound and meaning.” While various definitions and subtypes of sound symbolism have now been proposed in the literature (see, for instance, Hinton et al., 2006; Lockwood & Dingemanse, 2015; Perniss et al., 2010), this paper broadly construes sound symbolism to be “systematic associations between sounds and meanings.” As we will observe, however, the types of sound symbolism that we explore in the current paper demonstrably have their bases in the way we manipulate our speech, and we set aside those kinds of patterns that seem to have been established purely by convention, like English phonesthemes (e.g., *gl-* in English: Bergen, 2004, see footnote 2).

One major aim of the current study is to expand the empirical scope of sound symbolism by reporting new case studies. One of the best-known cases of sound symbolism is that, in many languages, the vowel [a] is associated with images that are larger than those connoted by [i] (e.g., Berlin, 1994, 2006; Jespersen, 1922; Sapir, 1929; Shinohara & Kawahara, 2016; Ultan, 1978). For example, Sapir (1929) showed that the nonce word *mal* – which contains the vowel [a] – is judged by English speakers to be a better word for a big table, whereas the nonce word *mil* – containing the vowel [i] – is judged to be a better name for a small table. Jespersen (1922) and Sapir (1929) are two pioneering studies on the sound symbolic values of different vowels in modern linguistic/psychological studies, and they both suggest that these sound-meaning associations may be grounded in the different articulatory configurations of these vowels; e.g., difference in oral aperture (how open the mouth is) or the volume of the suboral cavity in front of the

tongue.¹ This difference in articulatory configuration can be projected onto the semantic meanings of these vowels.² To quote Sapir (1929, p. 235):

[i]t is possible that the inherent ‘volume’ of certain vowels is greater than that of others and that this factor alone is sufficient to explain the results of the experiment. On the other hand, it should be noted that one may unconsciously feel that the tongue position for one vowel is symbolically ‘large’ as contrasted with the tongue position for another. In the case of *i* the tongue is high up toward the roof of the mouth and articulates pretty well forward. In other words, the vibrating column of air is passing through a narrow resonance chamber. In the case of *a* the tongue is very considerably lowered in comparison, and also retracted. In other words, the vibrating column of air is now passing through a much wider resonance chamber.

Ohala (1994) developed a general, acoustics-based theory of sound symbolism, now often referred to as the “Frequency Code Hypothesis.” This theory suggests that sounds with high frequency properties (either *f*₀ or *F*₂) can index an idea of smallness, because everything else being equal, sounds with high frequency energy imply a small vibrator or a small resonating cavity. Under this theory, [i] evokes images of smallness, because it has high *f*₀ and high *F*₂ – indeed, the sub-oral cavity in front of the tongue during the articulation of [i] is small, compared to the articulatory configurations of the other vowels (see Ohala, 1994: Figure 22.4 for illustration). Moreover, Shinohara & Kawahara (2016) have shown that *F*₂ is a good predictor of size ratings by speakers of Mandarin Chinese, English, Japanese and Korean.

1. While this experiment by Sapir (1929) is now generally considered as one of the major inspirations of modern experimental studies of sound symbolism, which is true to some extent, as an anonymous reviewer reminded us, his results may need to be interpreted with more caution. Sapir (1929) tested different comparisons among different vowels, and even tested those vowels that do not exist in English. The subsequent studies citing Sapir (1929) tend to focus on the clearest contrast – [a] vs. [i] – which may have exaggerated the importance of his finding. With these said, later studies following up on Sapir (1929) have established, convincingly in our opinion, that different vowels evoke different size ratings (patterns often referred to as “magnitude sound symbolism”: see e.g., Nuckolls, 1999).

2. Not all sound-meaning correspondences may have such a bodily basis in the articulatory nature of our speech. For example, the English *gl-* sequence is often used for words that mean something related to light (e.g., *glitter*, *glare* and *glow*), but there is no phonetic sense in which *gl-* emits any kind of light. As such, these patterns are sometimes referred to as “systematicity,” distinguished from other types of sound symbolism, and they are very often language-specific (e.g., Dingemanse et al. 2015). See Bergen (2004) for an extensive discussion of this pattern, also known as “phonaesthemes.” However, many if not all patterns of sound symbolism seem to have a basis either in articulatory or acoustic properties of the sounds (see Kawahara 2017, 2020 for recent exemplification), and the patterns that the current study analyzes are no exceptions.

Size-related sound symbolism is perhaps one of the best-known examples of sound symbolism, but later studies have revealed various other semantic dimensions for which sound symbolism can be relevant. Even if we limit ourselves to those images connotated by different vowel qualities, as Lockwood and Dingemanse (2015) succinctly summarize, the following semantic dimensions can be symbolically represented: precision, brightness, contour, taste, speed, size and redness/yellowness (see their Figure 1 for the references). The “multi-dimensionality” of sound symbolism raises a question regarding which semantic dimension can be (and cannot be) denoted by sound symbolism, a question that is important to address within the ever-growing body of research on sound symbolism. Our case studies reported below aim at addressing this general issue by exploring a new semantic dimension (“villain-ness”) which has been hitherto unaddressed. While we admit that our scope is modest, it is a stepping stone toward understanding the nature of sound symbolism in natural languages. Moreover, as we elaborate below, we will examine a set of sounds (=labial consonants) whose sound symbolic nature has been under-explored in the previous literature.

The Saussurean arbitrariness of linguistic signs generally had a substantial influence on modern thinking about languages. However, within cognitive linguistics (e.g., Johnson, 1987; Lakoff & Johnson, 1980, 1999; Langacker, 1987, 1991, 2008), sound symbolism has been considered to be an interesting topic of exploration. We believe that there are three reasons for why this has been the case. First, some instances of the non-arbitrary relationships between sounds and meanings can be understood as cases of *embodiment* (see Shinohara & Kawahara, 2016 and Yoshimura, 2014 for explicit discussion on this point). If the hypotheses entertained by Jespersen (1922), Sapir (1929) and Ohala (1994) reviewed above are on the right track, for example, it means that articulatory characteristics of different sounds, along with their acoustic consequences and perceptual qualities, can be projected onto the meanings of their sounds. The size of an oral aperture (or the size of the suboral cavity in front of the tongue) is mapped onto the size ratings of different vowel types. In other words, the experiences of producing and listening to sounds may have non-trivial influences on how we assign meanings to particular sounds (Perniss & Vigliocco, 2014). More generally speaking, some sound symbolic patterns show that our bodily experiences may shape the way we use our languages, which is one of the fundamental tenets of cognitive linguistics.

The second reason why sound symbolism attracts cognitive linguists' interests is the fact that some sound symbolic patterns may instantiate *iconicity* in natural languages (see in particular Dingemanse et al., 2015). For example, recent studies of Pokémon names, reviewed below in further detail, have revealed that Pokémon characters whose names are longer tend to be larger, heavier and/or stronger in Japanese (Kawahara et al., 2018b), English (Shih et al., 2018, 2019) and various

other languages including Cantonese, Korean and Russian (Shih et al., 2019). This systematic correspondence can be understood as a case of *the iconicity of quantity*, in which longer sounds represent something taller, larger and stronger (Haiman, 1980, 1985). This principle is observed outside of the Pokémon lexicon; we commonly observe that lengthening of a segment can be used to express a higher commitment to the proposition expressed by the speaker, as in “*it was sooooooooo weird*” or “*that flight was so loooooong*” (see e.g., Fuchs et al., 2019). To take another example, D’onofrio (2014) and Maurer et al. (2006) demonstrate that sounds produced with lips such as [b] and [u] tend to be associated with rounded figures. It is very likely, as these authors argue, that lip rounding gestures of these sounds are iconically mapped onto visual images of roundedness. As these examples illustrate, sound symbolism offers a window through which to examine the role of iconicity in natural languages, a topic that is actively explored in cognitive linguistics.

The third reason why sound symbolism may be interesting to study from the perspective of cognitive linguistics is the possibility that sound symbolism is an instance of a more general cognitive mechanism, namely, *cross-modal perception* in which sensation in one cognitive modality affects sensation in another modality (e.g. Lockwood & Dingemanse, 2015; Ramachandran & Hubbard 2001, Spence 2011). It has been found, for instance, that sounds have systematic relationships not only with meanings, but also with visual shapes (Köhler, 1947; see below for more) and taste (e.g., Crisinel & Spence 2009). The iconicity of quantity discussed above is also observed between other domains of cognition (Marks, 1978). A growing body of work shows that our cognitive system is organized in such a way that different modalities are actively influencing one another (Spence, 2011). Sound symbolism is no exception in the sense that it involves a cross-modal mapping from one modality (sounds/audition) to another modality (meaning/size/shape). These observations suggest that sound symbolism instantiates a case in which our linguistic knowledge is shaped by domain-general, rather than language-specific, cognitive mechanisms, supporting another tenet of cognitive linguistics.

2. Sound symbolism in proper names

For the reasons outlined above, sound symbolism is now a topic that is actively explored by phoneticians, psychologists, cognitive scientists, as well as by cognitive linguists. To contribute to this ever-growing body of literature, the present paper builds on research examining sound symbolic patterns in proper names, pursuing the general thesis that names are not chosen randomly. More specifically, we aim to provide further evidence for the hypothesis that certain types of

sound with particular phonological properties are chosen to capture an aspect of the named object or person.

More concretely, one major inspiration of our current research is a classic observation that male names and female names are characterized by different phonological features in languages like English and Japanese (Brown & Ford, 1961; Cassidy et al., 1999; Cutler et al., 1990; Slater & Feinman, 1985; Tessier, 2010; Whissell, 2001; Wright & Hay, 2002; Wright et al., 2005), and some of these features are demonstrably based on sound symbolic principles (see Sidhu & Pexman 2019 for a recent summary). For example, male names are more likely to contain obstruents (plosives, fricatives, affricates) than female names. In contrast, female names are more likely to contain sonorants (nasals, liquids, glides) than male names. These generalizations are known to hold both in English (Cassidy et al., 1999; Slater & Feinman, 1985; Tessier, 2010; Wright & Hay, 2002; Wright et al., 2005) and Japanese (Kawahara et al., 2015; Shinohara & Kawahara, 2013; Uemura, 1965). Beyond English and Japanese, cross-linguistically, less sonorous consonants are more likely to appear word-initially for a term meaning “father” than for a term meaning “mother” (e.g., [papa] vs. [mama]) (Murdock, 1959).

These generalizations are demonstrably rooted in another well-studied case of sound symbolism: the *takete-maluma* effect. Köhler (1947) demonstrated that the nonce word *takete* (containing only voiceless obstruents, [t] and [k]) is likely to be associated with an angular object, whereas the nonce word *maluma* (containing only sonorants, [m] and [l]) tends to be associated with a round object. This shape-sound association is replicated by a number of follow-up studies (e.g. D’Onofrio, 2014; Kawahara & Shinohara, 2013; Lindauer, 1990; Nobile, 2015). It is not too surprising that these sound-shape associations are extended to other semantic dimensions via further semantic associations; for example, prototypically, male bodies tend to be more angular than female bodies, and hence obstruents may have gotten to be associated with maleness mediated by the notion of angularity. Likewise, sonorants may have gotten to be associated with femaleness mediated by the notion of roundness.

Regarding the question of the phonetic basis of the *takete-maluma* effect, Kawahara & Shinohara (2012) conjectured that this sound-symbolic connection between obstruents and angular shapes derives from abrupt amplitude modulations during the burst and frication of these sounds, acoustic characteristics of obstruents. Jurafsky (2014) and Nobile (2015) entertain a similar idea. Aperiodic noise resulting from the rise of intraoral airpressure – defining features of obstruent – look, literally speaking, spikey and angular on waveforms (Figure 1(a), (b)). On the other hand, periodic energies of sonorants, reflecting regular vocal fold vibrations, look round on waveforms (Figure 1(c), (d)). These acoustic characteristics can be projected onto the perception of different types of shapes. To the

extent that this hypothesis is on the right track,³ these sound symbolic connections imply that listeners' auditory representations of the sounds, shaped by the experience of listening to their acoustic characteristics, non-trivially affect their linguistic knowledge (Perniss & Vigliocco, 2014).

Within this larger context of studies of sound symbolism, one emerging general project is exploration of sound symbolic patterns in the Pokémon characters' names, the project now known as Pokémonastics (Godoy et al., 2019; Kawahara et al., 2018a, 2018b; Kawahara & Kumagai, 2019a, b; Kumagai & Kawahara, 2019; Shih et al., 2018, 2019). As of 2017, there were more than 800 Pokémon characters, each possessing particular attributes such as weight, size, and strength. A growing body of work reveals that these attributes can be probabilistically predicted from the sound properties of their names. For example, voiced obstruents ([b], [d], [g], and [z]) were found to be associated with Pokémon characters that are larger, heavier and stronger in the Japanese Pokémon names (Kawahara et al., 2018b), and this sound symbolic association is demonstrably productive, as shown by experiments using new Pokémon characters and nonce names (Kawahara et al., 2018a; Kawahara & Kumagai, 2019a, b; Kumagai & Kawahara, 2019). Subsequent studies of the existing Japanese Pokémon names have shown in addition that bilabial consonants are associated with smaller characters (Shih et al., 2018, 2019).

As noted by Shih et al. (2018, 2019), using the Pokémon universe has some distinct virtues for studies of sound symbolism. First, this universe has a large number of creatures, which allows for statistical exploration of sound symbolic patterns in their names. Second, perhaps more importantly, the Pokémon universe has a fixed set of denotations (i.e., the Pokémon creatures), which is held constant across different languages. This feature is not what can be expected to hold in natural languages, because languages differ in terms of the set of denotations that they deploy. For example, Japanese lexically distinguishes “rice plant” (=ine), “cooked rice” (=gohan), and “raw rice” (=kome), but these distinctions are not present in English. Such cross-linguistic differences are pervasive; some languages do not even have a term for “left” or “right,” and instead use absolute direction terms, such as “east” or “west” (Levinson, 1997). If we were to use basic vocabulary to study sound symbolism across languages (e.g. Wichman et al., 2010), we would have to face this problem that the set under consideration is not consistent across languages. This problem does not arise in the Pokémon universe, because in any language we study, the set of characters that are named is

3. We fully admit that that establishing a causality relationship is impossible, and that Kawahara and Shinohara's (2012) proposal remains as a hypothesis.

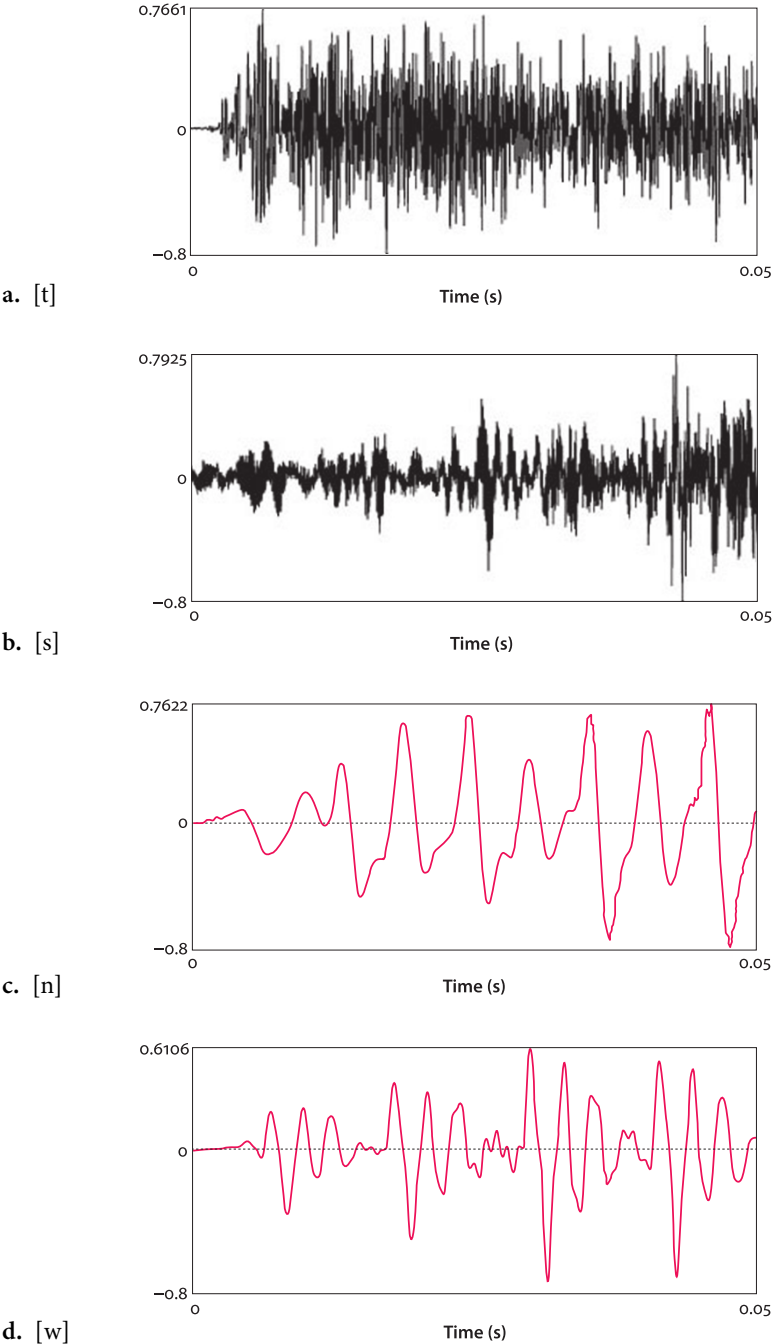


Figure 1. Waveforms of [t], [s], [n] and [w]. Obstruents (the top two panels) look angular, while sonorants (the bottom two panels) look round. Adapted from Kawahara (2017)

fixed.⁴ With these advantages in mind, one of the case studies reported below expands on the existing studies of sound symbolic patterns in Pokémon names.

2.1 The current study

The current study is a case study of sound symbolism in personal names, aiming to address one specific question: can villain-ness be symbolically expressed? As mentioned earlier, which semantic properties can be symbolically expressed in human languages is one important question that is (and most likely continues to be) addressed in the studies of sound symbolism. We test two specific hypotheses stated in (1). We will come back to the discussion of why we would expect villain-ness to be symbolically represented in the general discussion section (Section 4), but for now, we note that it is interesting to ask whether, in addition to very concrete concepts like size and shape discussed above, an abstract concept like “villain-ness” can be signaled by sound symbolism (see Auracher 2017 and Lupyan & Winter, 2018 for recent discussion on the relationship between sound symbolism and abstract concepts).

(1) *Hypotheses to be tested*

Hypothesis 1a. Voiced obstruents are favored in villains’ names.

Hypothesis 1b. Bilabial consonants are avoided in villains’ names.

Hypothesis (1a) is motivated by the observation that, in Japanese, voiced obstruents are generally associated with negative images (e.g., Hamano, 1986; Kawahara et al., 2008; Kubozono, 1999; Suzuki, 1962). For example, Suzuki (1962) discusses minimal pairs like *tori* (chicken) vs. *dori* (non-edible portion of chicken) and *kani* (crab) vs. *gani* (non-edible portion of crab), in which the initial voiced stops of the latter terms express non-edibility, something that is unfavorable. Kawahara (2017) reports that voiced obstruents appear much more frequently in monsters’ names in the *Urutoraman* TV series than in actual people’s names. Kawahara et al. (2008) report a nonce-word experiment showing that Japanese speakers judged voiced obstruents, especially those in initial syllables, to be dirty.

While this sound-symbolic association is psychologically very salient for Japanese speakers (and hence well-known in the literature), it is yet to be examined whether the same association holds for speakers of other languages. To the best of our knowledge, Shinohara and Kawahara (2009) is the only study which directly addressed this question; they presented pairs of pictures of the same

4. This methodology is foreshadowed by Berlin (1994, 2006) who studied the names of tinamou and rail (i.e., “ethnozoological nomenclature”) across 17 languages. In his work too, the set of denotation is fixed across all the target languages.

object, one in its clean state and the other in its dirty state (e.g., a dirty dish vs. a clean dish). Along with these pictures, they presented nonce words containing voiced obstruents and those containing voiceless obstruents (e.g., [zabe] vs. [sape]). Their results showed that Mandarin Chinese and English speakers, as well as Japanese speakers, tend to associate nonce words containing voiced obstruents with “dirty” pictures.⁵

This association between voiced obstruents and dirtiness might arguably have a bodily basis, in that the production of voiced obstruents involves a non-trivial aerodynamic challenge (e.g., Hayes & Steriade, 2004; Jaeger, 1978; Ohala, 1983; Proctor et al., 2010; Westbury & Keating, 1986). The oral constriction required to produce an obstruent blocks the airflow in the oral cavity, which raises the intra-oral air pressure, making it hard to sustain sufficient transglottal air pressure to maintain vocal fold vibration. Speakers thus need to resort to extra articulatory maneuvers to expand their supralaryngeal cavity, such as raising of the velum and lowering of the larynx, in order to accommodate this aerodynamic challenge. Indeed, some languages such as Hawaiian entirely lack voiced obstruents (Pukui & Elbert, 1979), and it is likely that this gap is due to this aerodynamic challenge (Hayes & Steriade, 2004). The aerodynamic challenge of voiced obstruents, because of the extra articulatory burden, may cause negative images for voiced obstruents.⁶ As the aerodynamic challenge of voiced obstruents is a matter of

5. A number of studies, starting with Newman (1933), have shown that English speakers associate voiced obstruents with images of largeness and/or heaviness. The analysis of English Pokémon names shows that there is a correlation between size/power and the number of voiced obstruents (Shih et al. 2019). One of the experiments reported by Kawahara and Breiss (2020) shows that English speakers tend to associate nonce names with voiced obstruents with larger, post-evolution Pokémon characters (though see Kawahara & Moore, 2020). Shih and Rudin (2019) demonstrate that when they examine the nicknames of Major League Baseball players, there is a positive correlation between the number of voiced obstruents and their height. Iwasaki et al. (2007) show that English speakers are at least partially sensitive to the meanings of voicing in obstruents when guessing the meanings of Japanese onomatopoeic expressions.

6. An underlying assumption here is that articulatory challenge, which we take to be a negative thing, can be projected onto negative images. As an anonymous reviewer pointed out, this is a big assumption which should not be taken for granted. The reviewer also suggested an alternative hypothesis: to quote, “since voiced obstruents are associated with larger size, this idea of larger size might itself be suggestive of negativity, since what is viewed in a negative way might seem larger and out of control.” One way to address this alternative is to test whether other sounds that are judged to be large (e.g., [a]) are also judged to be negative; testing this alternative is beyond the scope of the current paper, however. The reviewer also asked if we should consider this sound symbolic relationship to be a conventional one that is specific to Japanese, just like *gl-* sequence in English. We doubt that this is the case, as the effects of voiced obstruents are observed in languages other than Japanese, as our current study shows (see also Newman 1933; Iwasaki et al., 2007; Kawahara & Breiss, 2020; Shih & Rudin, 2019), and that this

aerodynamics (i.e., physics), it would come as no surprise if this sound-symbolic association were to be observed across languages, hence the Hypothesis (1a).

The first case study presented in the current paper tests Hypothesis (1a) by examining the names of English Disney characters. The prediction is that if voiced obstruents are associated with general negative images in English (just as in Japanese), voiced obstruents should be more likely to appear in the names of villains than in those of non-villains. To examine whether a similar tendency holds in Japanese, the second case study uses existing Japanese Pokémon names to investigate whether the names of villainous characters are more likely to contain voiced obstruents than those of non-villainous characters.

Hypothesis (1b) was inspired by the recent observation by Kumagai and Kawahara (2020) that bilabial sounds such as [p] and [m] are often used in the names of Japanese diapers (e.g., *mamiipoko* and *meriizu*). These studies show that not only are [p] and [m] often used in existing diaper names, when Japanese speakers are asked to come up with new diaper names and adult cosmetic names, all types of bilabial consonants appear in diaper names more frequently than in adult cosmetic names. Their follow-up forced-choice task using nonce words moreover shows that when presented with nonce words with bilabial consonants and those with coronal consonants, the former is more likely to be chosen for diaper names. In short, it appears to be the case that bilabials in Japanese can symbolically represent baby diapers.

Kumagai & Kawahara (2020) point out that bilabials emerge relatively early in the language acquisition process – bilabials occur very frequently in babbling, and they are one of the first sets of sounds that are acquired by infants in general (Jakobson, 1968; MacNeilage et al., 1997). Ota (2015) provides a summary of studies on the acquisition of different phonemes by Japanese children, and these studies generally show that bilabials are indeed among those that are acquired early.⁷ To the extent that bilabial consonants are very often used by babies, it is not too surprising that they are symbolically associated with the image of “baby-ness” (for a related proposal, see also Jakobson, 1969). Indeed, Japanese employs the onomatopoeic expression [babuu] to mimic babies’ babbling. Furthermore, Kumagai (2019) demonstrates that bilabial sounds such as [p] and [m] are judged to be cuter than any other consonants by Japanese speakers. In addition, Shih et al. (2018, 2019)

sound symbolic association arguably has an articulatory basis. However, we acknowledge that it is important to examine languages other than Japanese and English, if we are to argue that the sound symbolic connection is universal.

7. The bilabial fricative [ɸ] is a clear exception to this generalization. To reconcile this observation with their experimental results, Kumagai and Kawahara (2020) propose that their results instantiate a case of feature-based generalization; i.e., the sound-symbolic association holds at the level of distinctive features rather than at individual segments, involving some sort of abstraction.

found that Japanese Pokémon characters whose names include bilabial sounds tend to be smaller, which implies that Japanese speakers may make a sound-symbolic association between bilabial sounds and smallness. To summarize, the results of these studies predict that bilabial sounds would be avoided in villains' names, assuming that villainous characters are not meant to be cute, weak, or a baby-like.

Before we proceed, one caveat on our analyses is in order. As pointed out by an anonymous reviewer, since [b] is both bilabial and a voiced obstruent, the two hypotheses in (1) make conflicting predictions regarding [b]. This issue is inevitable, as a sound is defined on multiple dimensions (generally speaking, place of articulation, manner of articulation and laryngeal contrasts), each of which can have a sound symbolic value (see e.g., D'Onofrio, 2014 for multiple-dimensionality of the sounds, and what that means for sound symbolism). The same sound – for the case at hand, [b] – can potentially be both villainous and a baby-like (implying non-villain-ness) at the same time. However, one could be concerned that counting [b] in the analyses as representing both villain-ness and cuteness may promote confirmation biases. To be conservative, therefore, we excluded [b] from the analyses altogether.⁸

3. Names of villainous Disney characters

The first case study targeted Disney characters' names in English, and we believe that this is the first attempt to analyze Disney characters' names from the perspective of sound symbolism. Moreover, as far as we know, neither of the hypotheses in (1) has been tested using English names in general. The corpus of English Disney character names provides a sufficiently large dataset for quantitative testing of these hypotheses. Testing villains' names using Disney characters' names is interesting because the tension or conflict between heroes and villains is a prominent theme in Disney stories, and Disney designers (consciously or unconsciously) may therefore consider it to be important to highlight villains' evil qualities.⁹ This is not to deny that "surprise" is also an important factor in Disney films, and a character's villainous status should not be guessed solely on the basis of their names.

8. This analytical strategy is conservative to the extent that follow-up experimental studies show that [b] is suited to express both villain-ness (Kawahara & Kumagai, 2019b) and cuteness (Kumagai 2019; Kumagai & Kawahara, 2019b, 2020). Including [b] in the current analyses did not change the results.

9. See also Teshigawara (2003), who showed that villains' voices have some salient phonetic characteristics, demonstrably involving pharyngeal expansion and accompanying lowering of F2. See also Miyakoda and Oshita, 2019.

3.1 Method

Disney characters’ names were collected from “The Disney Wiki: Characters from theatrical animated features,” and villains’ names from “The Disney Wiki: List of Disney Villains.” Names labelled as “redeemed,” “neutral,” or “deleted” were excluded from the current analysis, as were titles such as “Mr.” and suffixes based on existing names (e.g., “king”), phrases containing prepositions (e.g., “of”) and determiners (e.g., “the”). For example, in the case of “Willie the Whale”, only the “Willie” portion was coded. If a particular name appeared in more than one movie, that name was counted only once. This procedure resulted in 179 villains’ names containing 297 obstruents and 785 names of non-villains containing 1,244 obstruents. Orthographic representations were automatically converted to semi-phonemic transcriptions using the LOGIOS Lexicon Tool, allowing us to access the presence of voiced obstruents and bilabial consonants in each name; the set of bilabial consonants in English included [p, m, w] (recall that [b] was excluded from the analysis). The set of names that was analyzed is made available online as a part of the supplementary materials at Appendix 1 and 2.

3.2 Results

Table 1 shows the distribution of voiced and voiceless obstruents in the names of Disney villains and non-villains. We observe that villains are more likely to have names that contain voiced obstruents (e.g., *Drizella*, *Gantu* and *Gaston*) than non-villains, and this skewed distribution is statistically significant ($\chi^2(1) = 9.868$, $p < .01$, $\phi = 0.80$). As expected, the sound-symbolic effect is not deterministic – that is, it is not a dichotomous decision but it instead manifests itself as a stochastic (probabilistic) tendency. This result suggests that while a language is a system that can connect sounds and meanings in an arbitrary way, as Saussure (1916/1972) and Hockett (1959) suggested, sound symbolic principles can exert their influence in a stochastic way.

Table 1. Voiced obstruents vs. voiceless obstruents in the names of Disney characters

	Voiced	Voiceless	Total
Villains	118 (39.7%)	179 (60.3%)	297
Non-villains	374 (30.1%)	870 (87.1%)	1,244
Total	492	1,049	1,541

Table 2 shows that bilabial sounds are more often used in the names of non-villains (e.g., *Pinocchio*, *Pocahontas*, *Moana*, *Mickey*) than in villains’ names. Although the size of this difference is small, it is nevertheless statistically significant ($\chi^2(1) = 3.877$,

$p < .05, \phi = 0.037$).¹⁰ This finding is especially interesting because, to the best of our knowledge, this sound symbolic value of bilabial consonants in English has been understudied. Descriptively, we can conclude that villains' names in English avoid bilabial consonants to a statistically significant degree; this pattern holds, maybe because English speakers (like Japanese speakers) associate bilabial consonants with images of innocence, cuteness or baby-ness, as these sounds are prototypically observed in babies' speech, as detailed in section 1.3.

Table 2. Bilabial consonants vs. non-bilabial consonants in the names of Disney characters

	Bilabial	Non-bilabial	Total
Villains	54 (9.7%)	500 (90.3%)	554
Non-villains	293 (12.9%)	1,974 (87%)	2,267
Total	347	2,474	2,821

4. Names of villainous Pokémon characters

The previous section has shown that it is possible to represent villain-ness with sound symbolism. Two questions arise: (1) is villain-ness expressed symbolically in a language other than English? and (2) would we observe a similar sound symbolic pattern in a domain other than Disney characters? To address these questions, we explore another empirical domain, the Pokémon universe. The analysis offered in this section was inspired by Pokémonastics, a series of studies on sound symbolic patterns in Pokémon names, which show different types of sound symbolic patterns in the Pokémon universe across different languages (see Section 1.2). These previous studies have targeted Pokémon characters' weight, height, evolution levels and their strengths and examined whether these parameters can be (probabilistically) predicted from their names. The present analysis explores one unexplored aspect of Pokémon characters, namely, *types*.¹¹ This study thus not only addresses the question of whether Japanese expresses villain-ness by way of sound symbolism, but it also expands the scope of the Pokémonastics research paradigm.

10. Since the two hypotheses in (1) were pre-planned rather than post-hoc, the alpha level was not corrected.

11. While this paper was under review, a follow-up experimental study further examined the productivity of the sound symbolic effects on Pokémon types (Kawahara & Kumagai 2019b). It still remains to be the case that this paper is the first one that has established the sound symbolic effects on Pokémon types.

4.1 Method

All Pokémon names available as of February 2018 (i.e., up to the 7th generation) were included in this analysis. In the games, Pokémon characters are classified into different types. In the current study, three villainous types were grouped together: Dark, Poison and Ghost. The aim was to explore whether these names were more likely to contain voiced obstruents and less likely to contain bilabial sounds, compared to other types of Pokémon characters. Those that are both Ghost and Fairy types were excluded because of their ambivalent status. Pokémon names that include symbols indicating gender (♂ and ♀) or numerical expressions were also excluded, as it is not unambiguously clear how these are to be read. This procedure yielded 141 villain names containing 277 obstruents, and 653 non-villain names containing 1,168 obstruents. Geminate consonants (i.e., long consonants) were counted as one instance, not two. Japanese contrasts plain (non-palatalized) consonants and palatalized consonants; both plain and palatalized consonants were counted. The set of target Japanese bilabial consonants thus included [p^(j), m^(j), φ^(j), w] ([φ] is a voiceless bilabial fricative; [w] does not have a palatalized counterpart). The list of names that was analyzed is made available as a part of the online supplementary materials.

4.2 Results

Table 3 shows the distribution of voiced obstruents and voiceless obstruents. As predicted, there are more voiced obstruents in villains’ names (e.g., *Abusoru*, *Dogaasu*, *Derubiru*) than in those of non-villains, and this difference is statistically significant ($\chi^2(1)=10.827$, $p<.001$, $\phi=0.087$). This result is compatible with the observation that Japanese speakers generally find voiced obstruents to be associated with negative images. It is also compatible with the observation by Kawahara (2017) that voiced obstruents very often appear in Japanese monsters’ names.

Table 3. Voiced obstruents vs. voiceless obstruents in the names of Pokémon characters

	Voiced	Voiceless	Total
Villains	111 (40.1%)	166 (59.9%)	277
Non-villains	346 (29.6%)	822 (70.4%)	1,168
Total	457	988	1,445

Table 4 shows the distribution of bilabial and non-bilabial consonants. Although villains’ names are less likely to contain bilabials than those of non-villains, the difference was not significant ($\chi^2(1)=2.922$, *n.s.*, $\phi=0.033$). To analyze the pattern in

further detail, we compared the Dark type (which in Japanese literally means “evil”) and the Fairy type, which is the least villainous. Each of these types had 46 names. As shown in Table 5, this follow-up analysis highlights the difference in the use of bilabial consonants between the two groups, which is significant ($\chi^2(1)=14.326$, $p<.001$, $\phi=0.216$). This finding accords well with the previous finding that bilabial consonants are associated in Japanese with images of babies (Kumagai & Kawahara, 2020). As bilabial consonants connote innocence, smallness and cuteness, they are not considered to be very appropriate for Dark characters.

Table 4. Bilabial vs. non-bilabial consonants in the names of Pokémon characters

	Bilabial	Non-bilabial	Total
Villains	66 (13.6%)	420 (86.4%)	486
Non-villains	359 (16.9%)	1,768 (83.1%)	2,127
Total	425	2,188	2,613

Table 5. Bilabial vs. non-bilabial consonants: Dark vs. Fairy

	Bilabial	Non-bilabial	Total
Dark	17 (10.7%)	142 (89.3%)	159
Fairy	42 (28.4%)	106 (71.6%)	148
Total	59	248	307

5. General discussion and conclusions

This paper has contributed to studies of sound symbolism, the stochastic and systematic correspondence between sounds and meanings, by showing that a concept as abstract as “villain-ness” can be signaled symbolically. Previous studies have shown that proper names of particular types are often characterized by particular phonological characteristics, some of which are based on sound-symbolic principles. Building on these studies, we have demonstrated that the names of both English Disney villains and Japanese Pokémon villains are more likely to contain voiced obstruents and less likely to contain bilabial sounds than those of non-villains (although the effect of bilabiality in Pokémon names was confined to a comparison between Dark and Fairy types). These novel findings augment the growing body of research confirming the existence of sound symbolism in natural languages. The effects of bilabial consonants in Disney names were especially interesting, as the sound symbolic effect of bilabial consonants is, to the best of our knowledge, understudied in English.

As discussed in Section 1.3, voiced obstruents may be associated with negative images because of the aerodynamic challenge they present, and extra articulatory adjustments that are called for to accommodate that challenge. Bilabial consonants may be associated with “baby-ness” because these are the prototypical sounds that babies most often make in their babbling. If these hypotheses are on the right track, it seems reasonable to hypothesize that the sound symbolic relationships that we have identified in the current studies may have their basis in the ways we manipulate and observe our speech. In that sense, then, sound symbolism constitutes a case of *embodiment* in language.

Finally, we would like to reflect briefly upon the further implications of the present study. While most previous studies of sound symbolism have investigated primary perceptual properties like size, shape, or hardness/softness, the present findings suggest that more abstract concepts like *evilness* may be symbolically associated with particular sets of sounds, which might yield important clues about the origins of ethics. Not all abstract concepts are probably signaled by sound symbolism; to our knowledge, there is no empirical evidence that the concept of “responsibility,” for example, can be symbolically associated with any phonological category. Where does this difference lie? We tentatively propose that, to survive, our human ancestors had to make informed judgments about other members of the society, especially in order to distinguish those who are not altruistic. As many researchers, including Tomasello (2009), have argued, humans’ altruistic tendencies distinguish them from other animals. Acquired through evolution, this trait is likely to have contributed to our survival. To sustain strong communities, we must be able to detect and eliminate “betrayers” – free-riders who exploit others’ altruism or tyrants who destroy the welfare of community members (Boehm, 2012). As proposed by Alexander (1987), the selection-by-reputation theory contends that humans may have evolved a sensitivity to society’s “bad members” or “evil ones” and learned to disseminate this information by means of language. If so, properties like *evilness* may not be totally abstract, but instead provide near-perceptual information to which we pay special attention. If sounds can signal who is likely to be vicious, this may also be advantageous, explaining at least in part why there can be an association between certain sounds and the image of *evilness*. See also the recent claim by Auracher (2017) who demonstrates that sound symbolism can signal notions related to social dominance.

This explanation finds an interesting parallel with what Shih et al. (2019) concludes based on an extensive cross-linguistic comparison of Pokémon names. They observe that unlike in the real world (see Section 1.2), we do not observe robust sound symbolic effects to signal gender differences in the Pokémon world. They argue that this difference arises because finding a mate is crucial for survival in the real world, but not so much in the Pokémon world. This claim is further

supported by the fact that Pokémon strength status is actively signaled by way of sound symbolism across languages, and this is so because Pokémon characters routinely fight with each other. They conclude that sound symbolism is actively deployed to signal those attributes that are important for their lives in the given world. Integrating their conclusions with ours, we would like to close this paper by suggesting that exploring the sound symbolism of ethics-related concepts may prove to be a fruitful topic for future studies, especially in the context of cognitive linguistics.

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Appendix 1. Disney characters’ names

Villains		Non villains					
grimhilde	hayabusa	geppetto	slue	gramps	esmeralda	jebidiah	krei
monstro	sabor	apatosaurus	macbadger	evinrude	djali	allardyce	mochi
chernabog	carnotaurus	parasaurolophus	ichabod	faloo	clopin	kashekim	bogo
ceratosaur	velociraptors	gryposaurus	tessel	marahute	madellaine	nedakh	clawhauser
ronno	yzma	diplodocus	baltus	vixey	philoctetes	lilo	fru
aconcagua	tiberius	ankylosaurus	jaq	taran	megara	pelekai	moana
tetti	gantu	corythosaurus	mome	eilonwy	hephaestus	pleakley	heihei
tatti	scroop	ornithomimus	raths	gurgi	fa	nani	pua
drizella	longbourne	brachiosaurus	ii	fflewddur	mulan	kawena	tui
maleficent	fayvoon	edmontosaurus	scamp	fflam	mushu	hamsterviel	sina
cruella	grewnge	psittacosaurus	dachsie	dallben	cri	yuki	gramma
vil	blinko	compsognathus	pongo	doli	shang	puloki	tala
badun	krailoni	chasmosaurus	tibs	eidilleg	terk	hasagawa	
kaa	mertock	plateosaurus	pellinore	flaversham	tantor	koda	
woozles	crex	pteranodon	scullery	mousetoria	kala	tuke	
orddu	dor	tylosaurus	mowgli	attina	kerchak	tanana	

Villains		Non villains				
orwen	gothel	nothosaurus	baloo	adella	plio	framagucci
orgoch	yama	dimorphodon	bagheera	andrina	yar	krunklehorn
ratigan	kakamora	plesiosaurus	hathi	aquata	zini	tiana
jetsam	tamatoa	archaeopteryx	raksha	lumiere	suri	bouff
mcleach		placochelys	akela	enchantress	neera	naveen
lefou		kannemeyeria	flunkey	nala	baylene	odie
d'arque		pegasi	ranjan	pumbaa	eema	atilla
jafar		brudus	messua	mufasa	url	vanellope
gazeem		centaurettes	bonfamille	rafiki	kuzco	schweetz
shenzi		jacchus	frou	sarabi	pacha	fluggerbutter
banzai		elephanchine	frou	zazu	chicha	jubileena
kovu		faline	roquefort	kiara	chaca	anon
zira		gurri	hautcourt	meeko	yupi	pabbie
vitani		panchito	tigger	powhatan	vinny	bulda
nuka		gaucioto	eeyore	nakoma	santorini	sitron
frollo		yasha	kanga	kocoum	gaetan	baymax
cerberus		pecos	roo	kekata	rocio	wasabi

Appendix 2. Pokémon characters' names

Villains	Non villains				Dark	Fairy
gengaa	aakeosu	genosekuto	dojotchi	perushian	sukatanku	ashireenu
goosu	aaken	kemusso	dodaitosu	peroppafu	sukanpuu	aburii
goosuto	aamarudo	kerarappa	dokkoraa	beroberuto	dorapion	aburibon
sukatanku	aianto	kerudio	dodekabashi	peroriimu	mikaruge	erufuun
sukanpuu	agirudaa	keromatsu	dotekkotsu	beroringa	yamirami	kapukokeko
dorapion	agunomu	kentarosu	todoguraa	houou	akujikingu	kaputetefu
mikaruge	agehanto	kenhorou	todozeruga	hoeruoo	abusoru	kapubururu
yamirami	agojimushi	koaruhii	toranseru	hoeruko	iberutaru	kapurehire
aabo	asanan	koikingu	toridepusu	hoohoo	gaogaen	kyuwawaa
aabokku	ashimari	koiru	toritodon	boomanda	kamanero	kiruria
akujikingu	ashireenu	goukazaru	torimian	pokabu	kibania	kuchiito
abusoru	azumaou	googooto	doryuuzu	bosugodora	kirikizan	guranburu
ariadosu	achamo	kootasu	torunerosu	pottaishi	guraena	kureffy
itomaru	anopusu	goorikii	doredia	potchama	gekkouga	saanaito

Villains	Non villains				Dark	Fairy
iberutaru	abagoora	kokodora	dorobanko	poppo	konohana	shushupu
utsudon	aburii	kokoromori	toropiusu	poniita	komatana	zeruneasu
utsubotto	aburibon	kojofuu	donfan	popokko	goronda	dianshii
utsuroido	amaajo	kojondo	donmeru	porigon	sazandora	dedenne
en-nyuuto	amakaji	kosumoumu	naetoru	horuudo	samehadaa	togekissu
oorotto	amamaiko	kosumoggu	nageki	borukenion	shizarigaa	togechikku
gaogaen	amarusu	kosokumushi	nagetsukesaru	borutorosu	jiheddo	togepii
kagebouzu	amaruruga	kodakku	nakkuraa	horubii	zuruzukin	ninfia
karamanero	ametama	gochimiru	nasshii	powarun	zuruggu	nemashu
kibania	amemoosu	gochimu	nattorei	mainan	zoroa	bariyaado
giratina	arigeitsu	gochizeru	namakero	magiana	zoroaaku	pii
kirikizan	aruseusu	kodora	namakobushi	magukarugo	daakurai	pikushii
girugarudo	annoon	gonyonyo	namazun	makunoshita	daatengu	pippi
kusaihana	iibui	kobaruon	nyaasu	magumaggu	choroneko	pukurin
kuzumoo	ishizumai	kofuurai	nyaonikusu	magumarashi	derubiru	pupurin
guraena	ishitsubute	kofukimushi	nyasupaa	makenkani	donkarasu	furaajesu
guregguru	inomuu	gomazou	nyabii	masheedo	nyuura	furaette
kurobatto	irumiize	komoruu	nyahiito	masukippa	nokutasu	furabebe
gekkouga	iwaaku	koratta	nyarumaa	maggyo	barujiina	purin
kokuun	iwaparesu	korinku	nyorozo	masshibuun	baruchai	buruu
gokurin	iwanko	gorudakku	nyorotono	massuguma	bangirasu	furefuwan
konohana	uindi	goroonya	nyorobon	manafi	burakkii	peroppafu
gobitto	wooguru	goroon	nyoromo	manene	herugaa	peroriimu
komatana	usokkii	korotokku	ninfia	mafokushii	pochiena	magiana
goruugu	usohachi	korobooshi	nuikoguma	mamanbou	maaiika	masheedo
gorubatto	udeppou	koromori	nuoo	mamepato	manyuura	manene
goronda	upaa	gonbe	numakuroo	mayurudo	meguroko	mariru
konpan	urimuu	saanaito	numeiru	marakatchi	monozu	mariruri
sazandora	urugamosu	saidon	numera	mariru	yamikarasu	mereshii
samayooru	eaamudo	saihoon	numerugon	mariruri	reparudasu	monmen
samehadaa	eipamu	sakurabisu	neiti	marumain	warubiaru	rarutosu
shizarigaa	eefi	sanagirasu	neitio	mankii	warubiru	ruriri
jiheddo	eteboosu	saniigo	neoranto	mantain		
shandera	eneko	sabonea	nekurozuma	manmuu		
junaipaa	enekororo	sawamura	nekkooara	mijumaru		
jupetta	ebiwaraa	zanguusu	nemashu	mizugorou		

Villains	Non villains				Dark	Fairy
shirodesuna	emuritto	sandaa	nendooru	mitsuhanii		
sukorupi	emonga	sandaasu	nokotchi	miniryuu		
sunabaa	erikiteru	sando	nozupasu	minezumi		
zubatto	erufuun	sandopan	haaderia	minomadamu		
supiaa	erureido	shiidora	paaruru	minomutchi		
subomii	erekiddo	jiiransu	baibanira	mimiroppu		
zuruzukin	erekiburu	sheimi	pauwau	mimiroru		
zuruggu	erezaado	sherudaa	baokkii	myuu		
zoroa	erebuu	shikijika	baoppu	myuutsuu		
zoroaaku	entei	jiguzaguma	haganeeru	mirutanku		
daakurai	enbuoo	jijiiron	hagigishiri	miruhoggu		
daatengu	enperuto	shishiko	bakuuda	mirokarosu		
dasutodasu	oosubame	shizukumo	bakuongu	muurando		
dadarin	oodairu	jibakoiru	bakugamesu	mukubaado		
tamagetake	ootachi	shibishirasu	bakufuun	mukuhooku		
choroneko	oobemu	shibibiiru	hakuryuu	mushaana		
desukaan	okutan	shibirudon	bashaamo	muchuuru		
desumasu	okorizaru	shimama	hasuburero	mukkuru		
derubiru	oshamari	janobii	hasuboo	munna		
dogaasu	otachi	jarako	basurao	meekuru		
dokukurage	otamaro	jararanga	batafurii	meganiumu		
dokukeiru	odoshishi	jarango	bachuru	metagurosu		
dokuroggu	onigoori	jarooda	pachirisu	metamon		
dohidoide	onishizukumo	shawaazu	hassamu	metangu		
doramidoro	onisuzume	jukain	patchiiru	meteno		
donkarasu	onidoriru	jugon	baffuron	mebukijika		
nazonokusa	ononokusu	shushupu	hatooboo	meraruba		
nidangiru	onondo	shubarugo	baniputchi	meriipu		
nidokingu	omusutaa	juputoru	baniritchi	mereshii		
nidokuin	omunaito	jiraachi	hanekko	meroetta		
nidoriina	onbaan	shiruvardi	banebuu	moukazaru		
nidoriino	onbatto	shinboraa	hahakomori	moguryuu		
nyuura	gaadi	suikun	hapinasu	mokuroo		
nukenin	gaameiru	zugaidosu	hayashigame	mokoko		
nokutasu	kaiooga	sutaamii	parasu	mojanbo		
baketcha	kairikii	sutoraiku	parasekuto	monjara		

Villains	Non villains				Dark	Fairy
habuneeku	kairyyuu	subame	hariteyama	monmen		
hariisen	kairosu	suriipaa	hariboogu	yajiron		
barujiina	kaenjishi	suriipu	harimaron	yadokingu		
baruchai	kakureon	suwanna	bariyaado	yadoran		
bangirasu	kachikooru	zekuromu	parukia	yadon		
panpujin	gachigorasu	zenigame	barukii	yanakkii		
biidoru	gabaito	zeburaika	parushen	yanappu		
hidoide	kabarudon	zeruneasu	barubiito	yayakoma		
hitotsuki	kabigon	serebii	hanteeru	yarukimono		
hitomoshi	kapukokeko	soonano	banbadoro	yareyuutan		
fuupa	kaputetefu	soonansu	pii	yanguusu		
fushigisou	kapubururu	sorugareo	biikuin	yanchamu		
fushigidane	kapurehire	sorurokku	biidaru	yan-yanma		
fushigibana	kabuto	daikenki	hiidoran	yukikaburi		
fushide	kabutopusu	dainoozu	pikachuu	yukinooo		
burakkii	gaburiasu	taipu nuru	pikushii	yukiwarashi		
pururiru	gaburumo	gagutorio	bikutini	yukushii		
burungeru	kapoeraa	dageki	hikozaru	yuniran		
fuwaraido	gamagaru	tattsuu	pijotto	yureidoru		
fuwante	gamageroge	tatsubei	pijon	yungeraa		
betobetaa	kamitsurugi	tatetopusu	pichuu	yoogirasu		
betobeton	kameeru	taneboo	bippa	yooterii		
herugaa	kamekkusu	daburan	pippi	yorunozuku		
pendoraa	kametete	tabunne	hitokage	yowashi		
hoiiga	gamenodesu	tamazarashi	hitodeman	raikou		
bokuree	kamonegi	tamatama	hinoarashi	raichuu		
pochiena	karakara	tamanta	hinoyakoma	raiboruto		
maaiika	garagara	darumakka	hihidaruma	raguraaji		
maashadoo	karasarisu	dangoro	bibiyon	rakurai		
madatsubomi	karanakushi	danbaru	biburaaba	rakkii		
matadogasu	karikiri	cherimu	hipopotasu	ratta		
manyuura	garuura	cherinbo	himanattsu	ratiasu		
marunoomu	gantoru	chigorasu	himeguma	ratiosu		
muuma	giaru	chikoriita	hiyakkii	rabukasu		
muumaaji	gigaiaasu	chaaremu	hiyappu	rapurasu		

Villains	Non villains				Dark	Fairy
meguroko	gigiaru	chaobuu	birijion	ramuparudo		
menokurage	gigigiaru	churine	biriridama	rarantesu		
monozu	kiteruguma	chobomaki	hinbasu	rarutosu		
morufon	kinogassa	chonchii	pinpuku	rankurusu		
morobareru	kinokoko	chiraamii	faiaroo	rantaan		
yatoumori	kibago	chirachiino	faiyaa	randorosu		
yabukuron	kimawari	chiriin	fione	riishan		
yamikarasu	kimori	chirutarisu	buizeru	riifia		
yukimenoko	kyatapii	chirutto	buusutaa	rioru		
yonowaaru	kyamome	tsutaaja	huudin	riguree		
yomawaru	gyaradosu	tsuchinin	buubaa	rizaado		
rafureshia	gyaroppu	tsutsukera	buubaan	rizaadon		
ranpuraa	kyuukon	tsubotsubo	buupiggu	ririira		
runa-aara	kyuremu	tsunbeaa	ferooche	ringuma		
reparudasu	kyuwawaa	diaruga	fokko	ruujura		
rozureido	kirinriki	dianshii	foretosu	rukario		
rozeria	kiruria	diguda	fukamaru	rugarugan		
rotomu	kireihana	teerunaa	fukusuroo	rugia		
warubiaru	kingudora	deokishisu	pukurin	rukushio		
warubiru	kinguraa	dekaguusu	futachimaru	ruchaburu		
	kuitaran	tekkaguya	putera	runatoon		
	gusokumusha	tekkarin	bunyatto	ruriri		
	kuchiito	tesshiido	bubii	runpappa		
	kunugidama	teppouo	pupurin	reakoiru		
	kumashun	dedenne	furaajesu	rejjaisu		
	guraadon	terakion	furaigon	rejigigasu		
	guraion	deribaado	furaette	rejisuchiru		
	guraigaa	denjumoku	purasuru	reshiramu		
	kurabu	denjimushi	furabebe	rejirokku		
	guranburu	denchura	furiizaa	rekkuuza		
	kurimugan	denryuu	furiijio	redian		
	kurumayu	dootakun	burigaron	rediba		
	kurumiru	doodoo	purin	rentoraa		
	guraishia	doodorio	buruu	roobushin		
	kureseria	dooburu	furefuwan	rokon		

Villains	Non villains				Dark	Fairy
	kureffy	doomiraa	furoozeru	wakashamo		
	kurebeesu	togekissu	burosutaa	washiboshi		
	kuwaganon	togechikku	purotooga	watakkō		
	keikouo	togedemaru	heigani	waninoko		
	keeshii	togepii	beiriifu	wanrikii		
	kekenkani	dogoomu	herakurosu			
	gekogashira	dosaidon	perappu			
	kekkingu	tosakinto	perippaa			

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